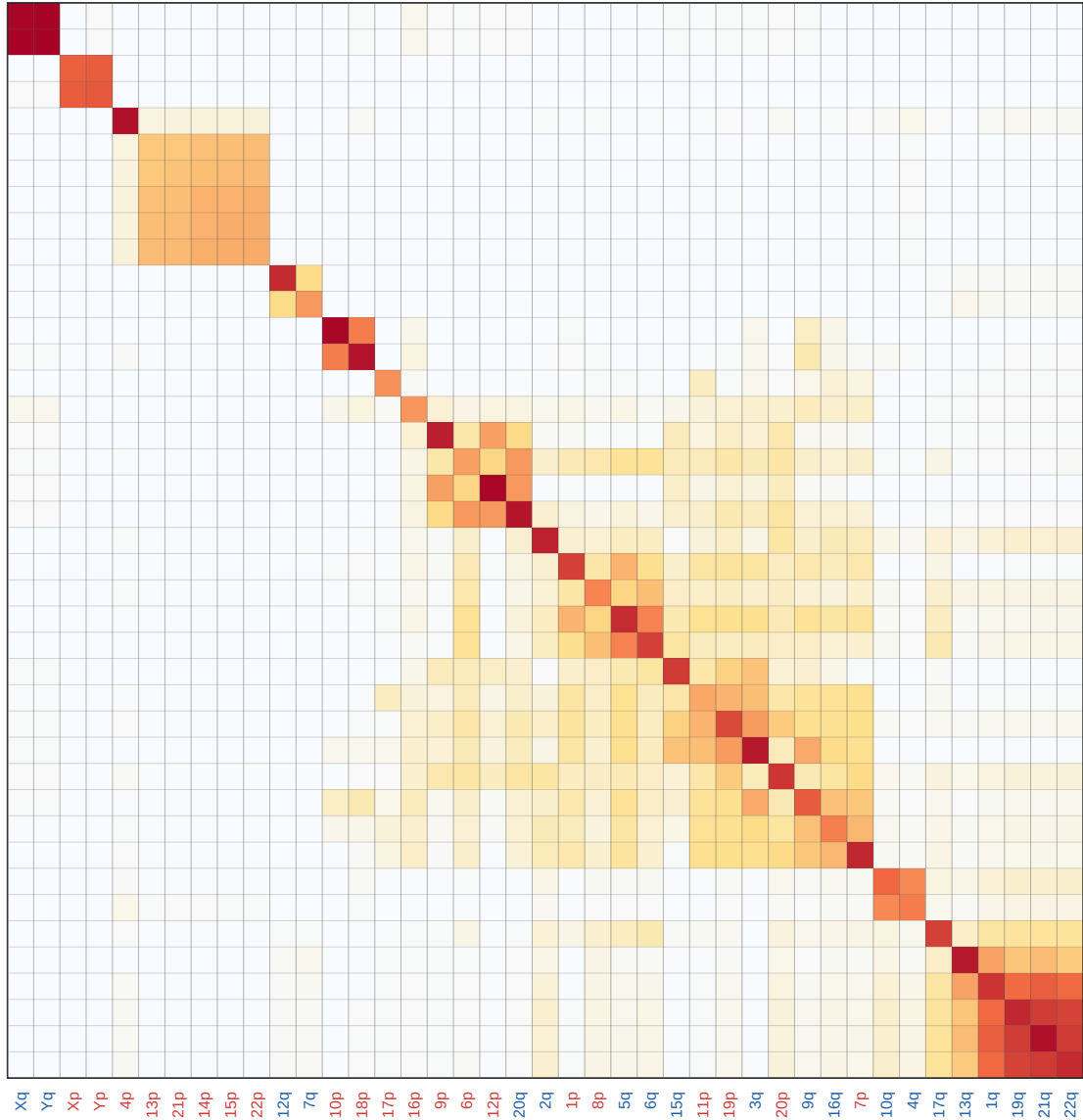
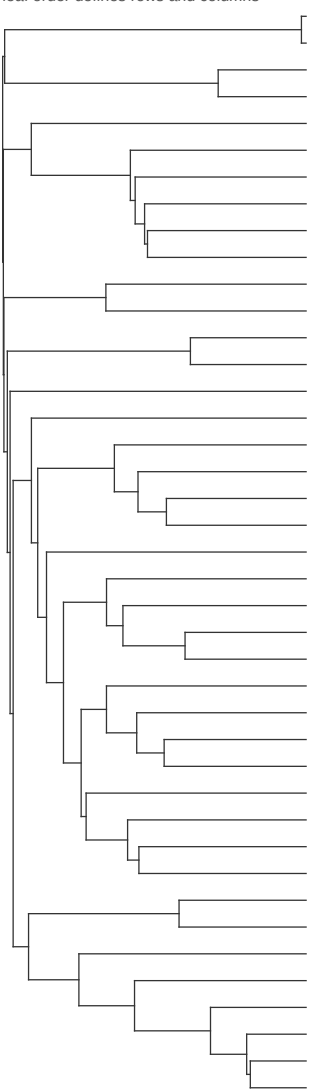


Slide 07a: UPGMA-aligned 41-arm heatmap

Tree tips, heatmap rows, and columns share one UPGMA order; labels: p = red, q = blue.

UPGMA tree
leaf order defines rows and columns



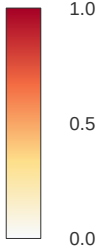
arms

- Xq
- Yq
- Xp
- Yp
- 4p
- 13p
- 21p
- 14p
- 15p
- 22p
- 12q
- 7q
- 10p
- 18p
- 17p
- 16p
- 9p
- 6p
- 12p
- 20q
- 2q
- 1p
- 8p
- 5q
- 6q
- 15q
- 11p
- 19p
- 3q
- 20p
- 9q
- 16q
- 7p
- 10q
- 4q
- 17q
- 13q
- 1q
- 19q
- 21q
- 22q

Encoding

- p arm label
- q arm label

Jaccard similarity



PDF uses vector cells/tree.
PNG is rendered directly at 4800x2700.
No stretched raster crop.